

Q2.2 The Writing Component (2 marks)

Using initial state 3, 3, 0, 0, L:

1. Run UCS with Cost Model A and Cost Model B, report:

- The solution of Q2.1 (UCS, cost model A) is:
- Solution Path: (3, 3, 0, 0, L) -> (3, 1, 0, 2, R) -> (3, 2, 0, 1, L) -> (3, 0, 0, 3, R) -> (3, 1, 0, 2, L) -> (1, 1, 2, 2, R) -> (2, 2, 1, 1, L) -> (0, 2, 3, 1, R) -> (0, 3, 3, 0, L) -> (0, 1, 3, 2, R) -> (0, 2, 3, 1, L) -> (0, 0, 3, 3, R)
- Total cost = 23
- Number of node expansions = 14

- The solution of Q2.1 (UCS, cost model B) is:
- Solution Path: (3, 3, 0, 0, L) -> (3, 1, 0, 2, R) -> (3, 2, 0, 1, L) -> (3, 0, 0, 3, R) -> (3, 1, 0, 2, L) -> (1, 1, 2, 2, R) -> (2, 2, 1, 1, L) -> (0, 2, 3, 1, R) -> (0, 3, 3, 0, L) -> (0, 1, 3, 2, R) -> (0, 2, 3, 1, L) -> (0, 0, 3, 3, R)
- Total cost = 17
- Number of node expansions = 14

2. Compare with BFS:

- Does UCS choose a different sequence of actions?
For our initial state (3,3,0,0,L), BFS, UCS Cost Model A, and UCS Cost Model B all returned the same sequence of states.

- How does total cost (UCS) relate to path length (BFS)?
BFS and UCS returned the same 11-action paths, but BFS assigned a total cost of 11 because each action is treated as one step, while UCS Cost Model A assigned it a total cost of 23, because each missionary costs 2 units and each cannibal costs 1 unit. UCS Cost Model B also returned the same 11-action path, and had a total cost of 17, because left-to-right trips cost 2 and right-to-left trips cost 1. Path length and path cost are different measures when actions have non-uniform costs, so despite the fact that all three searches returned the same path for this problem, it doesn't mean BFS will always minimize non-uniform path cost.

- Which cost model leads to more/fewer node expansions?
The different cost functions changed the evaluated cost of the solution, but the number of expanded nodes stayed the same. BFS and both UCS models still expanded 14 nodes

3. In 1–2 paragraphs, explain why UCS is necessary with non-uniform costs, and discuss the trade-off between least cost and fewest steps.

Uniform Cost Search is necessary when actions have non-uniform costs because the solution with the fewest actions might not be the solution with the lowest total cost. BFS expands states according to their depth, so it minimizes the number of actions when each action is treated

equally. UCS on the other hand prioritizes states according to their accumulated path cost ($g(n)$) and will choose a path containing more actions if those actions result in a lower overall cost.

It just so happened that for this problem BFS, UCS Cost Model A, and UCS Cost Model B all returned the same path, but they still optimized different factors. Cost Model A produced a total cost of 23 and Cost Model B produced a total cost of 17, even though both used the same 11-action path.

Q3.2 The Writing Component

Part A — Admissibility Proofs (6 marks)

For each heuristic (h_1 , h_2):

Heuristic 1: $h_1(s) = 2M_{\text{left}} + C_{\text{left}}$

h_1 is admissible. Under Cost Model A, every missionary remaining on the left bank eventually crosses to the right at least once, costing 2 units per missionary, and every cannibal remaining on the left eventually crosses for 1 unit per cannibal. $2M_{\text{left}} + C_{\text{left}}$ represents a lower bound on the remaining cost. The heuristic ignores additional costs caused by returning the boat and passengers making extra crossings, so it might underestimate the true cost but it can't overestimate it, so $h_1(s) \leq h^*(s)$ meaning h_1 is admissible.

For the initial state, $h_1(s) = 2(3) + 3 = 9$ The actual optimal cost found by UCS is 23, so h_1 is an optimistic estimate for the initial state.

Heuristic 2: $h_2(s) = \left\lceil \frac{2m_{\text{left}} + C_{\text{left}}}{3} \right\rceil$

h_2 is also admissible. Since $2M_{\text{left}} + C_{\text{left}}$ is nonnegative

$\left\lceil \frac{2m_{\text{left}} + C_{\text{left}}}{3} \right\rceil \leq 2M_{\text{left}} + C_{\text{left}}$ for the possible integer values in this problem.

This means, $h_2(s) \leq h_1(s)$ and since h_1 is already a lower bound on the true remaining cost, $h_2(s) \leq h_1(s) \leq h^*(s)$, which means h_2 can't overestimate the optimal remaining cost and is admissible.

For the initial state: $h_2(s) = \left\lceil \frac{2(3) + 3}{3} \right\rceil = 3$

Part B — Implementation and Comparison (8 marks)

1. Implement A* with h_1 and h_2 .
2. Run both on the initial state 3, 3, 0, 0, L
3. Record results in a table:

Heuristic	Solution Path (brief)	Total Cost	Node Expansions	Same Cost as UCS?
h1	3, 3, 0, 0, L) -> (3, 1, 0, 2, R) -> (3, 2, 0, 1, L) -> (3, 0, 0, 3, R) -> (3, 1, 0, 2, L) -> (1, 1, 2, 2, R) -> (2, 2, 1, 1, L) -> (0, 2, 3, 1, R) -> (0, 3, 3, 0, L) -> (0, 1, 3, 2, R) -> (0, 2, 3, 1, L) -> (0, 0, 3, 3, R)	23	13	Yes
h2	(3, 3, 0, 0, L) -> (3, 1, 0, 2, R) -> (3, 2, 0, 1, L) -> (3, 0, 0, 3, R) -> (3, 1, 0, 2, L) -> (1, 1, 2, 2, R) -> (2, 2, 1, 1, L) -> (0, 2, 3, 1, R) -> (0, 3, 3, 0, L) -> (0, 1, 3, 2, R) -> (0, 2, 3, 1, L) -> (0, 0, 3, 3, R)	23	14	Yes
h3	(3, 3, 0, 0, L) -> (3, 1, 0, 2, R) -> (3, 2, 0, 1, L) -> (3, 0, 0, 3, R) -> (3, 1, 0, 2, L) -> (1, 1, 2, 2, R) -> (2, 2, 1, 1, L) -> (0, 2, 3, 1, R) -> (0, 3, 3, 0, L) -> (0, 1, 3, 2, R) -> (0, 2, 3, 1, L) -> (0, 0, 3, 3, R)	23	13	Yes

4. Compare node expansions: which heuristic was more efficient, and why?

h1 was more efficient than h2 for the initial state because A* with h1 expanded 13 nodes, while A* with h2 expanded 14 nodes. Both still returned an optimal cost of 23, which was the same cost as UCS, but h1 provides a tighter estimate of the remaining cost than h2 by giving A* more information about which states are promising. In comparison, the additional information allowed A* to avoid one expansion that occurred in using h2.

Part C — Designing a Tighter Heuristic (6 marks)

Propose your own heuristic h_3 that is admissible and which dominates h_1 or h_2 . Note, we are not looking for the $\max(h_1, h_2)$ heuristic. We want you to get creative with your heuristic.

I designed h_3 to improve upon h_1 by accounting for the position of the boat and unavoidable return crossings needed to transport everyone.

If $N = M_{\text{left}} + C_{\text{left}}$ is the number of people remaining on the left bank.

We get $R_{\text{Min}}(s) = \begin{cases} \max(0, N-2), & \text{if the boat is on the left} \\ N, & \text{if the boat is on the right and } N > 0 \end{cases}$

Giving us $h_3(s) = 2M_{\text{left}} + C_{\text{left}} + 2R_{\text{min}}(s)$

The main difference between h_3 and h_1 & h_2 is that h_3 incorporates the structure of the boat movement itself. h_1 estimates the minimum passenger cost of moving everyone remaining on the left across once, and h_2 creates a lower bound based on that remaining passenger weight, but neither of them account for where the boat currently is or for the boat needing to repeatedly return to transport more people. h_3 uses both the number of people remaining and the current side of the boat to estimate unavoidable extra travel.

For the initial state, $N = 3 + 3 = 6$ and the boat is on the left. This means $R_{\text{min}} = 6 - 2 = 4$
Which gives us $h_3 = 2(3) + 3 + 2(4) = 17$

Why h_3 is admissible:

$2M_{\text{left}} + C_{\text{left}}$ accounts for the minimum cost of moving every person currently on the left to the right at least once; however, the boat can carry up to two people. After crossing, the boat needs to return whenever more people still need transportation. The additional term $2R_{\text{min}}$ uses the cheapest possible extra movement. Under Cost Model A, a cannibal has a cost of 1, and a passenger who returns the boat to the left must eventually cross to the right again, leaving us with at least $1 + 1 = 2$ additional units of travel. The heuristic counts only this minimum extra cost and ignores circumstances in which a missionary must return or additional movements are necessary, so it remains a lower-bound estimate rather than intentionally adding the full true remaining cost.

Why h_3 dominates h_1 :

$h_3(s) = h_1(s) + 2R_{\text{min}}(s)$. Since $R_{\text{min}}(s) \geq 0$, we get $2R_{\text{min}}(s) \geq 0$

Therefore $h_3(s) \geq h_1(s)$, so h_3 dominates h_1 while incorporating additional information about the problem.